An Investigation into Paradoxes Between Latent Aptitude and Observed Statistics via Novel Full House Modeling

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This is a hopefully self-contained writeup that contains the foundational material needed to understand what results we need to move this work forward.

Sentences in bold text will denote results that we need investigated.

Sentences in red text will denote attempts that we have made.

1 Introduction

The following paragraph introduces the intuition behind Full House Modeling. It is adapted from **?** with some slight changes:

Our contribution (to discussion of baseball) is a novel modeling approach that extends Gould [1996]'s "full house of variation" concept about observed statistics from some system[1] to the broader source population of individuals eligible for selection into this system. This method works by using an estimate of the size of the source pool as a modeling input and connecting order statistics of observed statistics from individuals in the system under study to individuals in the source population. With distributional assumptions placed on the process that assigns latent talent to individuals in the source population and a pairing between observed statistics and latent aptitude such that the best performer according to a particular metric is paired with the highest latent aptitude, the second best performer is paired with the second highest latent aptitude and so on, we can estimate the latent aptitude of all individuals in the system. These mechanics allow for a balancing between how far one stands from their peers and the size of the source population. To have a high aptitude score, one must stand out from the peers in their system, and be a product of a large source population. However, this does not eliminate high achievers from smaller source populations from the top ranks.

[1]Stephen Jay Gould was a highly influential paleontologist, evolutionary biologist, and historian of science whose conceptual ideas on performance in baseball, rooted in evolutionary biology, have been widely recognized. However, his ideas remained conceptual and were not formalized mathematically. Our work codifies Gould's intuition into a principled mathematical framework grounded in statistical ideas.

2 Model Setting

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying latent $\mathbb{R}$-valued aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{iid}{\sim} F_X$. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j. We then define an indicator function $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right)$ as a determined and known system inclusion function for population i where a 1 indicates inclusion and a 0 indicates exclusion. We now suppose that an individual j included in system i has an observed $\mathbb{R}$-valued statistic $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$, $n_i \leq N_i$. As we will see later, the observed $Y_{i,j}$ values are one-to-one realizations of $X_{i,j}$, provided that $g_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1$. Therefore $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$.

Let $X_{i,(j)}$ and $Y_{i,(j)}$ denote the jth order statistic for system i. Here, larger numbers will correspond to a notion of "better." Thus $X_{i,(j)} > X_{i,(k)}$ whenever $j > k$, for example. We now form (Y, X) pairs of random variables for each individual included in the system. These pairs are of the form $(Y_{i,(j)}, X_{i,(r_{i,j})})$ where $r_{i,j}$ indexes the aptitude of the jth highest value observed from the system. As an example, let $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$, then the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs take the form $(Y_{i,(j)}, X_{i,(N_i - n_i + j)})$ for $j = 1, \ldots, n_i$.

The key Full House Modeling is the explicit connection between the elements forming the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs. Full House Modeling states this connection as

$$\begin{aligned} Y_{i,(j)} &= F^{-1}_{Y_{i,(j)}} \circ F_{X_{i,(r_{i,j})}}\left(X_{i,(r_{i,j})}\right), \\ X_{i,(r_{i,j})} &= F^{-1}_{X_{i,(r_{i,j})}} \circ F_{Y_{i,(j)}}\left(Y_{i,(j)}\right), \end{aligned} \tag{1}$$

where *circ* denotes functional composition, $1_i(X_{i,(r_{i,j})}, \mathbf{X}_{i,-(r_{i,j})}) = 1$, and both F_{Y_i} and F_X are continuous and invertible. We summarise Full House Modeling in the definition below.

Definition 2.1 (Full House Model)**.** *Consider populations with sizes N_i and system sizes n_i, $N_i \geq n_i$. Let a system inclusion function $1_i(\cdot, \cdot)$ be specified. Let $X_{i,j} \overset{iid}{\sim} F_X$, $j = 1, \ldots, N_i$. Let $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$. Form pairs $(Y_{i,(j)}, X_{i,(r_{i,j})})$, $j = 1, \ldots, n_i$ for all individuals such that $1_i(X_{i,(r_{i,j})}, \boldsymbol{X}_{i,-(r_{i,j})}) = 1$. Then the Full House Model generates $Y_{i,(j)}$ values following (1) where F_{Y_i} and F_X are continuous and invertible.*

Note that Definition 2.1 fixes F_X but allows F_{Y_i} to vary across systems.

3 Paradoxes Under Identical Distributions

In this section we will provide explicit paradox probability results in a Full House Modeling setting involving two populations with the additional specification that $F_{Y_i} = F_Y$, $i = 1, 2$ in Definition 2.1. The following definition formally defines the notion of paradox used throughout this article.

Definition 3.1. *Consider the Full House Model in Definition 2.1 with two populations denoted by $i = 1, 2$. We say a paradox has occurred when both $X_{1,(r_{1,j})} > X_{2,(r_{2,k})}$ and $Y_{1,(j)} < Y_{2,(k)}$ have occurred. The paradox probability is defined as*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right). \tag{2}$$

As a simple illustrative example of the above, let $r_{i,t} = N_i - n_i + t$ and set $j = n_1$ and $k = n_2$, and further suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then the paradox probability (2) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ when $N_1 \to \infty$. We now introduce a convenient representation which allows the paradox probability to be expressed in terms of the latent aptitudes alone.

Lemma 3.2. *Consider the Full House Model in Definition 2.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Then,*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right), \qquad (3)$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$.

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}), f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{1,2,j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. □

Our first formal result provides conditions that allow for an explicit paradox probability result. Those conditions require the following definition.

Definition 3.3. *Let $\alpha, \beta > 0$. Then the regularized incomplete beta function is defined as*

$$I_{\alpha,\beta}(x) = \frac{1}{B(\alpha,\beta)} \int_0^x u^{\alpha-1}(1-u)^{\beta-1} du,$$

where $B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the beta coefficient.

Note that the regularized incomplete beta function is the cumulative distribution function of a Beta(α, β) distribution.

Theorem 3.4. *Consider the Full House Model in Definition 2.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Denote by I the regularized incomplete beta function. Further assume that*

$$I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \geq 0, \qquad \text{for all } t \in (0,1). \qquad (4)$$

Under these assumptions,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&= \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2-n_1-p+k+1)}{B(k, n_2-k+1)} - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2-N_1-p+r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}$$

Remarks:

1. The result of Theorem 3.4 is distribution free. It only depends on the sample sizes n_i, N_i, $i = 1, 2$, the system inclusion function, and the indexes of the order statistics under comparison. The distribution on Y is eliminated by Lemma 3.2. To see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F_Y^{-1} \circ U_{(j)}$ where $U_{(j)} \sim \text{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = I_{j,n_1-j+1} \circ F_Y(y)$. With this in mind, from Lemma 3.2 we have that

$$\begin{aligned} h_{1,2,j,k}(x) &= F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(x) \\ &= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ \left(F_Y \circ F_Y^{-1}\right) \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x) \\ &= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x). \end{aligned}$$

   The above is the upper-bound of an integral with respect to Xs in the proof of Theorem 3.4. The influence of the distributional specification on Y does not influence the calculation of the paradox probability result in Theorem 3.4.

2. The distribution on X is eliminated when calculating the paradox probability (2) in its Lemma 3.2 representation, $\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right)$. These calculations involve double integrals with respect to densities for $F_{X_{1,(r_{1,j})}}$ and $F_{X_{2,(r_{2,k})}}$. One integral evaluation and a reparameterization remove the specification of F_X from the probability calculation as seen in the proof of Theorem 3.4.

3. The steps mentioned in the last sentence of the previous remark allow the probability calculation for $\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right)$ to simplify to

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt.$$

   The technical condition (4) ensures that this integral defines a non-negative probability in strong terms. We demonstrate with analytic and numeric arguments when (4) is satisfied and the results of Theorem 3.4 therefore hold.

Proof. First, Lemma 3.2 allows us to write

$$\text{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. We can rewrite $h_{1,2,j,k}(\cdot)$ into a form that will allow for specific computation involving Beta distributions. o see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F_Y^{-1} \circ U_{(j)}$ where $U_{(j)} \sim \text{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = g_{n,j} \circ F_Y(y)$, where we write $g_{n,j}(x) = I_{j,n-j+1}(x)$ for ease of notation. This expression of distribution functions allows us to write

$$\begin{aligned} &\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right) \\ &= \int_{-\infty}^{+\infty} \left\{ \int_x^{F_X^{-1} \circ g^{-1}_{N_1,r_{1,j}} \circ g_{n_1,j} \circ g^{-1}_{n_2,k} \circ g_{N_2,r_{2,k}} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \end{aligned}$$

$$
\begin{aligned}
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1}\circ g_{N_1,r_{1,j}}^{-1}\circ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x)} d\ g_{N_1,r_{1,j}}(F_X(y)) \right\}\ f_{X_{2,(r_{2,k})}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x) - g_{N_1,r_{1,j}}\circ F_X(x) \right\}^+\ f_{X_{2,(r_{2,k})}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x) - g_{N_1,r_{1,j}}\circ F_X(x) \right\}^+\ d\ g_{N_2,r_{2,k}}(F_X(x)) \\
&= \int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^+\ dt.
\end{aligned}
$$

Condition (4) states that

$$
\int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^+\ dt = \int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}\ dt.
$$

Thus,

$$
\begin{aligned}
&\int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^+\ dt \\
&\quad = \int_0^1 g_{n_1,j}(x)\ dg_{n_2,k}(x) - \int_0^1 g_{N_1,r_{1,j}}(x)\ dg_{N_2,r_{2,k}}(x) \\
&\quad = \int_0^1 I_x(j, n_1-j+1)\frac{x^{k-1}(1-x)^{n_2-k}}{B(k,n_2-k+1)}\ dx \\
&\qquad - \int_0^1 I_x(r_{1,j}, N_1-r_{1,j}+1)\frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(k,N_2-k+1)}\ dx \\
&\quad = \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p(1-x)^{n_1-p}\frac{x^{k-1}(1-x)^{n_2-k}}{B(k,n_2-k+1)}\ dx \\
&\qquad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p(1-x)^{N_1-p}\frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k},N_2-r_{2,k}+1)}\ dx \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2+n_1-p-k+1)}{B(j,n_2-k+1)} \\
&\qquad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k},N_2-r_{2,k}+1)}.
\end{aligned}
$$

□

Often it is of interest to compare the individuals with the highest aptitudes in their respective populations. Below we state the paradox probability in such setting using a Full House Model with two populations with the specification that $F_{Y_1} = F_Y$ for $i = 1, 2$.

Corollary 3.4.1. *Suppose that the conditions of Theorem 3.4 hold. Suppose that* $1_1(X_{1,(N_1)}, \boldsymbol{X}_{1,-(N_1)}) = 1_2(X_{2,(N_2)}, \boldsymbol{X}_{2,-(N_2)}) = 1$. *Then,*

$$
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \le \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.
$$

A similar result holds for the individuals with the lowest latent aptitudes in the population.

Corollary 3.4.2. *Suppose that the conditions of Theorem 3.4 hold. Suppose that* $1_1(X_{1,(1)}, \boldsymbol{X}_{1,-(1)}) = 1_2(X_{2,(1)}, \boldsymbol{X}_{2,-(1)}) = 1$. *Then,*

$$\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \begin{cases} \frac{N_2}{N_1+N_2} - \frac{n_2}{n_1+n_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.$$

We now demonstrate a setting in which technical condition (4) holds. This setting is based on a configuration of n_j, N_j, $j = 1, 2$ under the assumption that the systems are populated with individuals from the source populations who have the highest aptitudes.

Proposition 3.5. *Consider the Full House Model in Definition 2.1 with two populations, where* $F_{Y_i} = F_Y$ *for* $i = 1, 2$, *and let the system inclusion function be* $1_i(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$. *Denote by* I *the regularized incomplete beta function. Suppose that* $\frac{n_1}{n_2} \leq 1$ *and* $\frac{N_2}{N_1} \leq 1$, *and set* $k = j$. *Under these assumptions, we conclude that technical condition (4) holds and the conclusion of Theorem 3.4 applies.*

Proof. Specification of $1_i(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$ implies that $r_{i,j} = N_i + 1 - j$.

When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $x \in (0, 1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased [Askitis, 2021], or n_2 or N_1 are decreased. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned} & I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\ \implies & I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\ \implies & I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t. \end{aligned}$$

Technical condition (4) holds and the conclusion of Theorem 3.4 applies. □

We think that Proposition 3.5 can be generalized to $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$ with a different argument.

We tried the following in an attempt to generalize:

Let

$$f_{n_1,n_2,N_1,N_2,j}(x) = I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x).$$

Applications of the chain rule, the inverse function theorem [Rudin et al., 1964] on Page 221, and this property of regularized incomplete beta functions

$$I'_{\alpha,\beta}(I^{-1}_{\alpha,\beta}(x)) = \frac{I^{-1}_{\alpha,\beta}(x)^{\alpha-1}(1 - I^{-1}_{\alpha,\beta}(x))^{\beta-1}}{B(\alpha, \beta)},$$

allow us to write

$$
\begin{aligned}
f'_{n_1,n_2,N_1,N_2,j}(x) &= \frac{I'_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x)}{I'_{n_2-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x)} - \frac{I'_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x)}{I'_{N_2-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x)} \\
&= \frac{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-j-1}\left(1-I^{-1}_{n_2-j,j+1}(x)\right)^{j}/B(n_1-j,j+1)}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_2-j-1}\left(1-I^{-1}_{n_2-j,j+1}(x)\right)^{j}/B(n_2-j,j+1)} \\
&\quad - \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-j-1}\left(1-I^{-1}_{N_2-j,j+1}(x)\right)^{j}/B(N_1-j,j+1)}{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_2-j-1}\left(1-I^{-1}_{N_2-j,j+1}(x)\right)^{j}/B(N_2-j,j+1)} \\
&= \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)}\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2} - \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)}\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2} \\
&= \left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}\frac{B(N_2-j,j+1)}{B(N_1-j,j+1)} \\
&\quad \times \left[\frac{B(n_2-j,j+1)/B(n_1-j,j+1)}{B(N_2-j,j+1)/B(N_1-j,j+1)} - \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}\right]
\end{aligned}
$$

From here we see that $f_{n_1,n_2,N_1,N_2,j}(0) = f_{n_1,n_2,N_1,N_2,j}(1) = 0$, and when $x = 1$, we have

$$
\begin{aligned}
f'_{n_1,n_2,N_1,N_2,j}(1) &= \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)} - \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)} \\
&= \frac{(n_1-j)\ldots(n_1)}{(n_2-j)\ldots(n_2)} - \frac{(N_1-j)\ldots(N_1)}{(N_2-j)\ldots(N_2)} \\
&\leq 0,
\end{aligned}
$$

from the assumption that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$. If we can prove the $\frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}$ is a monotonic decreasing function for $x \in (0,1)$ and ranges from 0 to 1, then we can conclude that $f_{n_1,n_2,N_1,N_2,j}(x) \geq 0$ as desired.

References

Dimitris Askitis. Logarithmic concavity of the inverse incomplete beta function with respect to the first parameter. *Dimitris Askitis*, 2021.

Stephen Jay Gould. *Full House: The Spread of Excellence from Plato to Darwin*. Harvard University Press, 1996.

Walter Rudin et al. *Principles of mathematical analysis*, volume 3. McGraw-hill New York, 1964.